

Full Research Plan

Molecular Mechanisms of Milk Protein Adsorption at the Air-Water Interface

From Spontaneous Adsorption to Calcium Bridge and Fat Interaction

Force Field: CHARMM36m | Simulator: GROMACS 2020.4 | April 2026

1. Research Vision

This research program aims to establish the first complete atomistic description of milk foam stabilization — from individual protein behavior at the air-water interface to the cooperative effects of calcium bridging and fat interaction. The work is structured as four sequential phases, each producing independent publishable results while building toward a unified molecular model.

Phase	Title	System	Target Journal
Phase 1	BLG Adsorption (current)	1BEB + water slab	Langmuir / J. Phys. Chem. B
Phase 2	Casein Adsorption	α s1-Casein + water slab	Same paper as Phase 1
Phase 3	Calcium Bridge	BLG + Casein + Ca^{2+} at interface	Food Hydrocolloids
Phase 4	Fat Interaction	Proteins + Ca^{2+} + triglyceride	Nature Food / ACS Nano

2. Phase 1 — BLG Spontaneous Adsorption (Current)

2.1 Simulation Sets

Set 1A — Center of slab (completed 1,000 ns). Protein placed at the geometric center of the water slab (~3.5 nm from interface). This set provides the bulk diffusion baseline and has revealed a critical partial unfolding event at 600-800 ns.

Set 1B — Near interface / Plan B (in preparation). Protein CoM positioned 2.183 nm from the upper interface. Three independent replicas (500 ns each, gen_seed: 1234, 5678, 9999). Expected adsorption within 100-300 ns.

Set 1C — Placed at interface (existing data). Protein placed 2 Angstrom above water surface with hydrophobic face directed downward, 500 ns production run. Provides interfacial behavior and conformational change data.

2.2 Key Results from Set 1A (1,000 ns)

- Spontaneous adsorption from bulk center requires >1 μs — first atomistic quantification
- Partial unfolding event at 600-800 ns: R_g increased from 1.50 to 1.55 nm (irreversible)
- RMSD rose from 0.15 nm (0-400 ns) to 0.40 nm peak during unfolding event
- Partial unfolding temporally correlates with protein drift toward upper interface
- Loop BC (residues 30-35) identified as most flexible non-terminal region (RMSF ~0.53 nm)
- Hydrophobic SASA increased 20% over 1,000 ns — progressive pre-activation

2.3 Analysis Protocol

All three simulation sets will be analyzed using a standardized Python pipeline:

- Z-position tracking and distance to interface (track_z_position.py)
- RMSD analysis: backbone, beta-sheet, alpha-helix, hydrophobic patch (rmsd_analysis.py)
- SASA analysis: total, hydrophobic, hydrophilic, calyx region (sasa_analysis.py)
- Orientation analysis: hydrophobic patch vector and principal axis vs Z-axis (orientation_analysis.py)
- Advanced analysis: per-residue RMSF, Rg, conformational space (advanced_analysis.py)

3. Above-Water Protein Simulation — Scientific Value

Concept: Place the protein in the vacuum region above the water slab, approximately 0.5-1.0 nm above the air-water interface, and observe how it interacts with and penetrates the interface from the air side.

3.1 Why This Approach Has Scientific Merit

Although placing a protein in vacuum may seem unphysical at first glance, this setup can answer a fundamentally different and complementary scientific question:

Simulation Type	Scientific Question	What To Learn
Set 1A (bulk center)	Does protein diffuse to interface spontaneously?	Diffusion timescale, pre-activation mechanism
Set 1B (near interface)	What happens when protein approaches interface?	Adsorption pathway, orientation selection
Set 1C (placed at interface)	How does protein behave at interface?	Unfolding, footprint, surface pressure
Above-water (new)	Does protein adsorb from the air side? Is adsorption reversible?	Asymmetry of adsorption, desorption barrier, air-side vs water-side mechanism

3.2 Specific Scientific Questions Addressed

1. Adsorption asymmetry: Is adsorption from the air side thermodynamically different from the water side? Some surfactant proteins are known to adsorb preferentially from one phase.
2. Desorption barrier: Once adsorbed, can the protein desorb back into vacuum? This quantifies the energy well depth and foam stability.
3. Orientation from air side: Which face of BLG contacts the water surface first when approaching from above? This may differ from the water-side approach, revealing orientation-dependent adsorption mechanisms.
4. Structural integrity in vacuum: How does BLG behave in the absence of solvent? The protein will attempt to minimize exposed hydrophobic surface by collapsing — comparing this to the solvated state reveals the role of hydration in structure maintenance.

5. Interface penetration depth: How deeply does the protein penetrate the water surface from the air side? This provides the 'insertion depth' that is experimentally measurable by neutron reflectometry.

3.3 Proposed Setup

- Starting configuration: take final frame of Set 1B or 1C, translate protein upward by +5 to +8 nm above the water surface
- Protein CoM Z: ~26-29 nm (vacuum region, box Z = 35 nm)
- No position restraints, no pulling force — fully spontaneous
- Production: 200-500 ns (adsorption from above expected to be fast due to no diffusion barrier in vacuum)
- Temperature: 298 K with V-rescale, but note that vacuum region has no thermostat molecules — protein temperature will fluctuate more

3.4 Expected Observations and Publication Value

This simulation will likely show one of three scientifically interesting outcomes:

- Protein adsorbs rapidly from vacuum, penetrating deeply into the water surface — supporting amphiphilic collapse mechanism
- Protein hovers above interface, showing partial insertion with hydrophobic residues leading — demonstrating directional insertion
- Protein partially unfolds in vacuum before reaching water — revealing the role of hydration in structural stability

Any of these outcomes is publishable as it directly compares to Set 1A and 1B, completing a full picture of BLG adsorption from both phases. This comparison is novel and has not been reported in the literature to date.

Recommended as a supplementary simulation set to be included in Paper 1, adding a unique mechanistic perspective that distinguishes this work from all prior studies.

3.5 Technical Considerations

- Vacuum has no viscosity — protein will move very fast initially. Use a short timestep (1 fs) or position restrain for 1-2 ns to let the protein gently settle near the interface before releasing
- Monitor protein temperature separately — vacuum region thermostat artifact can cause artificial heating
- The water interface may deform significantly when a large protein lands on it — this is physically meaningful and should be measured as 'interface deformation depth'
- Compare with Set 1C (placed at interface from water side) to assess asymmetry

4. Phase 2 — Casein Adsorption and Comparison

4.1 Scientific Rationale

Caseins are intrinsically disordered proteins (IDPs) that constitute approximately 80% of milk protein. Unlike BLG with its stable beta-barrel structure, caseins lack persistent secondary structure and are expected to adsorb via a fundamentally different mechanism — rapid, disorder-driven interfacial attachment with minimal energy barrier.

Property	BLG (Phase 1)	α s1-Casein (Phase 2)
Structure	Folded beta-barrel	Intrinsically disordered (IDP)
MW	~18 kDa	~23 kDa
Adsorption speed	Slow (>1 μ s from bulk center)	Expected fast (<200 ns)
At interface	Partial unfolding required	Immediate reorganization
Ca ²⁺ binding	Surface Asp/Glu residues	Phosphoserine clusters (high affinity)
Structure source	PDB: 1BEB (X-ray, 1.8 Å)	AlphaFold2 model or NMR ensemble

4.2 Simulation Design

- System: α s1-Casein + water slab (12 x 12 x 35 nm, CHARMM36m, TIP3P)
- Set 2A: Near interface (1.5-2.0 nm) — 3 replicas x 500 ns
- Set 2B: Placed at interface — 500 ns (mirror of Set 1C for comparison)
- Additional analysis: End-to-end distance, Rg fluctuations, per-residue contact probability at interface

5. Phase 3 — Calcium Bridge Mechanism

5.1 Scientific Gap

While experimental studies demonstrate that Ca²⁺ improves foam stability in milk systems, the atomistic mechanism by which Ca²⁺ bridges adjacent protein molecules at the air-water interface remains completely unknown. No MD simulation study has reported Ca²⁺-mediated cross-linking between milk proteins at a slab interface.

5.2 Key Residues

BLG Ca²⁺ binding sites: D33, E44, D64, E89, D96, E108 (surface Asp/Glu residues)

Casein high-affinity sites: pSer41, pSer46, pSer48, pSer64, pSer66, pSer67 (phosphoserine cluster in α s1-Casein)

5.3 Simulation Systems

- System 3A: BLG + Ca²⁺ (5-10 mM) at interface — single protein Ca²⁺ binding

- System 3B: Casein + Ca²⁺ (5-10 mM) at interface — phosphoserine-Ca²⁺ dynamics
- System 3C: BLG + Casein + Ca²⁺ at interface — inter-protein bridge formation (main system)
- Box size may need to expand to 15 x 15 x 35 nm to accommodate two proteins

5.4 Analysis

- Ca²⁺ radial distribution function (RDF) around binding residue oxygens
- Ca²⁺ coordination number and residence time at each site
- Protein-protein distance and contact map at interface
- Interface film mechanical properties: surface tension and lateral diffusion
- Optional: PMF calculation via Umbrella Sampling for bridge formation free energy

6. Phase 4 — Fat Interaction and Complete Foam Model

6.1 Fat Molecules

Molecule	Type	Force Field	Scientific Role
Tripalmitin (PPP)	Saturated triglyceride	CHARMM36	Major milk fat — pro-foam
Triolein (OOO)	Unsaturated triglyceride	CHARMM36	Saturation comparison
Palmitic acid	Free fatty acid (sat.)	CHARMM36	Anti-foam agent
Oleic acid	Free fatty acid (unsat.)	CHARMM36	Anti-foam comparison

6.2 Key Questions

- Does Ca²⁺ bridging protect the protein film from displacement by free fatty acids?
- Which triglyceride saturation state is more disruptive to the protein film?
- What is the insertion depth of fatty acids into the protein-decorated interface?
- For large systems: consider MARTINI coarse-grained force field for computational efficiency

7. Timeline to Publication

Period	Activity	Deliverable	Status
Q2 2026 (Now)	Complete Plan B replicas, analyze Set 1A+1B+1C, write Paper 1 draft	Paper 1 manuscript draft	IN PROGRESS
Q3 2026	Phase 2: Casein simulations (2A + 2B), comparative analysis BLG vs Casein	Paper 1 submission	PLANNED
Q4 2026	Phase 3: Calcium Bridge systems (3A, 3B, 3C), bridge analysis	Paper 2 draft	PLANNED
Q1 2027	Paper 2 revision + Phase 4 setup (fat systems)	Paper 2 submission	PLANNED
Q2-Q3 2027	Phase 4 production runs and analysis	Paper 3 draft	PLANNED
Q4 2027	Paper 3 submission — complete foam model	3 papers published	GOAL

8. Computational Resource Estimate

Phase	Systems	Total simulation	GPU hours (est.)	Priority
Phase 1	3 sets + 3 replicas	~3,500 ns	~35,000 h	HIGH (now)
Phase 2	2 sets x 3 replicas	~3,000 ns	~30,000 h	HIGH (Q3)
Phase 3	3 systems x 500-1000 ns	~2,500 ns	~30,000 h	MEDIUM
Phase 4	4 systems x 500 ns	~2,000 ns	~25,000 h	MEDIUM
Total		~11,000 ns	~120,000 h	

9. Key References

The following papers provide the scientific foundation and comparative context for this research program.

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Analysis scripts: GitHub repository (in preparation) | Simulation data available upon request